Jaundel Simon

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Education

Toronto Metropolitan University — Toronto, ON
Bachelor of Engineering in Computer Engineering — *Sep 2023 – May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Advanced Computer Architecture, Embedded Systems Design, Systems-on-Chip Design, Computer Networks, Intelligent Systems

Technical Skills

Languages: Python, JavaScript, Java, C, VHDL, MATLAB
Machine Learning & Simulation: NumPy, PyTorch, TensorFlow.js, SciPy, scikit-learn, Three.js, WebGL 2
Hardware & Tools: FPGA/RTL, GHDL, Intel Quartus Prime, ModelSim/Questa, Git, GitHub Actions, Vite

Projects

Omni: Live Digital Twin | *Python, PyTorch, OpenCV, NumPy* — Sep 2026 – Present

- Leading a 4-person team building a live digital twin that reconstructs and renders physical workspaces in 3D from multiple RGB-D cameras, with persistent object identities, spatial memory, and scene-change detection
- Raised object-detection recall from 68.6% to 96.4% at 98.1% precision across 1,538 eligible object-view references from two development recordings
- Built perception and evaluation pipelines spanning YOLOE, MapAnything, MobileSAM and MediaPipe, using HO-Cap and BOP HOPE datasets for object detection, segmentation and 3D reconstruction

TMU MCP: AI Academic Workflow Integration | *Python, TypeScript, MCP, Cloudflare* — Oct 2026

- Built and deployed an MCP integration connecting AI assistants to D2L course information, assignments, announcements, Gmail, Google Drive, and Calendar to centralize routine academic information retrieval
- Implemented Python collectors and cached, normalized records with source references, incremental updates, and freshness indicators to make course documents, email, and calendar data available to AI clients
- Developed an OAuth-protected MCP endpoint on Cloudflare Workers with D1/KV storage and read-only retrieval tools; separated application access from private browser sessions and upstream credentials

SEVEN: Energy-Aware LLM Routing | *Python, Lemonade Server, OpenAI/Groq APIs* — Nov 2025

- Founded and led a 5-person team to build a router across local SLM, direct API, and cloud paths; won TMU Hack the World and a $1,625 prize
- Implemented Lemonade HTTP integration with device configuration, response validation, and exponential-backoff retries; modeled a 38% energy reduction under a 40% local-routing scenario
- Authored setup runbooks and managed pull requests and code reviews across the terminal UI, energy estimates, and conversation logging integration

MedMNIST from Scratch | *Python, NumPy, PyTorch, scikit-learn* — Sep 2025

- Generalized an 80-line NumPy MLP with handwritten forward/backpropagation to four 28×28 medical-image benchmarks, reaching 0.825 accuracy and 0.931 AUC on PneumoniaMNIST
- Implemented balanced metrics, saliency, occlusion, FGSM, cross-view shift, 5-client FedAvg, Forward-Forward, and pruning; retained 0.843 pneumonia accuracy after removing 90% of weights and fine-tuning

Explore more projects in my portfolio

Experience

Graphic Installer — Sep 2023 – Aug 2026
Signscapers Installation Inc. — *Greater Toronto Area*

- Installed wayfinding signs, branded wall graphics, privacy-frost vinyl, and point-of-purchase displays across client sites including the NBA, Scotiabank, TIFF, Microsoft, RBC, and Wine Rack
- Coordinated with client representatives and installation crews to confirm placement requirements, communicate site constraints, and address on-site questions throughout each installation